

MANAS AGARWAL

Ghaziabad, Uttar Pradesh

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EDUCATION

Dr. A.P.J. Abdul Kalam Technical University

Sept 2022 - Sept 2026

Bachelor of Technology in Computer Science and Information Technology (CSIT).

Senior Secondary School, CBSE , 92.2%

2022

Saraf Public School

EXPERIENCE

Educase India, Backend Intern

March 2025 – Present

- Optimized **SQL queries** and **database indexing**, reducing query execution time and improving API performance.
- Developed and tested RESTful APIs in Node.js, fixing **25+ critical bugs** and improving backend stability.
- Streamlined **API integration** with the frontend, improving response efficiency and reducing system downtime.

Programming Club, Coordinator

Feb 2024 – Present

- Developed a scalable Programming Club website with a team, reaching **5000+** viewership. [Programming Club](#)
- Worked and managed a team of **40+ members** to conduct various technical activities, doubt sessions, and coding competitions.
- Conducted a series of Data Structure and Algorithm workshops, named **#include** and mentored over 100+ students.

ACHIEVEMENTS

- Achieved **3rd Rank** in **DSA Coding Contest** with 350+ participants in **5th semester** organized by AKGEC, Ghaziabad.
- Achieved **4th Rank** in **DSA Coding Contest** with 300+ participants in **4th semester** organized by AKGEC, Ghaziabad.
- **Knight** at Leetcode [manas_agarwal16](#), **Pupil**, 1342 at Codeforces [manas_agarwal16](#), **3 Star** at Codechef [manasagarwal19](#)
- Solved **450+** questions on **Data Structures and Algorithm** and **500+** Questions on **Competitive Programming**.

PROJECTS

Chess Master | WebRTC, Socket.io, Sequelize ORM, PostgreSQL | [Github](#) | [Video Demo](#) | [Site Link](#)

- Leveraged **WebRTC** to let players engage in **live discussions** during gameplay, enabling them to share strategies, learn tricks, and recognize patterns for mastering chess, **serving over 100 players**.
- Empowers players to **revisit and analyze their past games**, providing insights into their best moves and misjudgments to enhance their performance.
- Developed an adaptive **rating system** starting at 1200, incorporating **mathematical and logical calculations** to accurately reflect player skill levels up to 2000.
- **Allows players to challenge their friends** and also ensures a secure sign-in and sign-up process through **OAuth integration**.

VStream | ReactJS, Redux Toolkit, NodeJS, Express, MongoDB | [Frontend](#) | [Backend](#) | [Site Link](#)

- VStream is a **unified video and music streaming** platform that offers users the convenience of watching videos and listening to music in one place.
- **MongoDB's Aggregation Pipelines** for 30% to 50% faster data processing and **Pagination** enabling 70% faster data fetching.
- Integrated **PayPal payment gateway** to enable secure transactions and premium upgrades.
- Implemented **JSON Web Tokens** for authentication and **OTP verification** for secure email validation.
- Built a responsive front-end with ReactJS, Redux Toolkit, and TailwindCSS for efficient state management and modern UI.

TECHNICAL SKILLS

Programming Skills: Proficient in Data Structures & Algorithms, with strong Competitive Programming experience

Programming Languages: C, C++, JavaScript, SQL, TypeScript

Technologies and Frameworks: ReactJS, Redux Toolkit, NodeJS, Express, Sequelize ORM, Prisma ORM

Databases and Cloud: MongoDB, MySQL, PostgreSQL, Amazon Web Services (AWS), Docker, Vercel, Render

Tools: Visual Studio Code, Git, Github, Github Desktop, Postman, etc.